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Table 1 Compilation of LD_{50} Values for Cyanide Compounds in Various Animal Species

Animal species	LD_{50} (mg/kg)	Molar dose (mmol/kg)	CN^- compound
Black vulture	4.80	0.098	NaCN
American kestrel	4.00	0.082	NaCN
Japanese quail	9.40	0.192	NaCN
Domestic chicken	21.0	0.428	NaCN
Eastern screech owl	8.60	0.175	NaCN
European starling	17.0	0.347	NaCN
Rat	5.72	0.117	NaCN
Mouse	8.50	0.131	KCN
Rat	9.69	0.149	KCN
Rabbit	5.82	0.089	KCN
Rat	3.62	0.134	HCN
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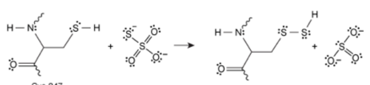
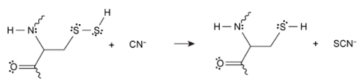
Based on animal models, one treatment developed for cyanide poisoning involves assisting the rhodanese enzyme in metabolizing CN^- to SCN^- via an intravenous solution of $NaNO_2$ and $Na_2S_2O_3$. The $Na_2S_2O_3$ assists in the formation of a reactive S–SH (persulfide) bond in the enzyme on cysteine-247 (Reaction 1), which then reacts with the cyanide ion to form thiocyanate (Reaction 2).



Reaction 1



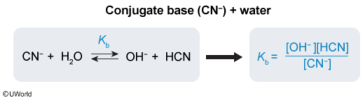
Reaction 2

Which of the following is the correct expression for the base dissociation constant K_b of CN^- in water?

- A. $\dfrac{[CN^-]}{[OH^-][HCN]}$ [21%]
- B. $\dfrac{[OH^-][HCN]}{[H_2O][CN^-]}$ [8%]
- C. $\dfrac{[H_3O^+][CN^-]}{[HCN]}$ [26%]
- ✔ D. $\dfrac{[OH^-][HCN]}{[CN^-]}$ [43%]

Correct
43% answered correctly

Explanation:



In solution, the acid ionization establishes an **equilibrium**, which can be described by the acid dissociation constant K_a. Similarly, the conjugate base formed during the ionization of an acid also interacts with water and establishes a second equilibrium, described by the **base dissociation constant K_b**. The K_a and K_b constants are defined as **molar concentration ratios** of the product species and the reactant species (omitting water).

$$K_x = \frac{[\text{products}]}{[\text{reactants}]}$$

Ionization of HCN in water occurs when HCN donates a proton to water to form hydronium (H_3O^+), which also forms a proton-accepting cyanide ion (CN^-) in the process.

$$HCN(aq) + H_2O(l) \rightleftarrows H_3O^+(aq) + CN^-(aq)$$

Accordingly, CN^- is the conjugate base of HCN, and the base dissociation occurs when CN^- accepts a proton from H_2O to form hydroxide (OH^-) and HCN.

$$CN^-(aq) + H_2O(l) \rightleftarrows OH^-(aq) + HCN(aq)$$

As such, the base dissociation constant K_b is expressed as:

$$K_b = \frac{[OH^-][HCN]}{[CN^-]}$$

(Choice A) The dissociation constants are defined to have the reaction products in the numerator and the reactants in the denominator of the ratio (ie, products over reactants). This expression for K_b is inverted.

(Choice B) The amount of solvent (water) exceeds the other chemical species by many orders of magnitude, and its concentration is relatively unchanged. For this reason, water should not be included in the K_b expression.

(Choice C) This is the *acid* dissociation constant K_a, not the *base* dissociation constant K_b.

Educational objective:
Acid ionization in an aqueous solution produces hydronium ions and a corresponding conjugate base of the ionized acid. Both the acid and its conjugate base interact with water and establish equilibria that can be described by molar concentration ratios in the form of the acid dissociation constant K_a and the base dissociation constant K_b.

Time spent: 0 sec
Subject: General Chemistry

QID: 401507
System: Solutions and Electrochemistry

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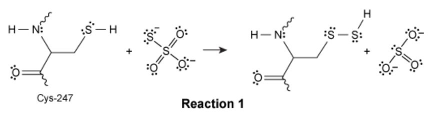
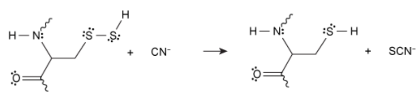


Reaction 1



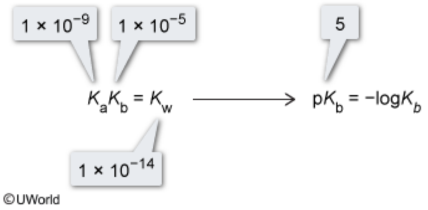
Reaction 2

The acid dissociation constant K_a for HCN is approximately 1×10^{-9}. What is the value of the pK_b?

- A. −5 [1%]
- B. 5 [85%]
- C. 9 [12%]
- D. 14

Correct
85% answered correctly

Explanation:



The **acid dissociation constant** K_a and the **base dissociation constant** K_b associated with the aqueous solution equilibria of an acid and its conjugate base are defined as molar concentration ratios of the product species and the reactant species (omitting water). These **ratios are related** to each other by the **self-ionization constant** K_w of water according to the following relationship:

$$K_a K_b = K_w$$

Because K_a is given as 1 × 10^{-9}, dividing K_w by the known K_a permits the calculation of K_b:

$$K_b = \frac{K_w}{K_a}$$

$$K_b = \frac{1.0 \times 10^{-14}}{1.0 \times 10^{-9}}$$

$$K_b = 1.0 \times 10^{-5}$$

Using this value determined for K_b, the pK_b can be determined:

$$pK_b = -\log K_b$$

$$pK_b = -\log\left(1.0 \times 10^{-5}\right)$$

$$pK_b = 5$$

Alternatively, a logarithmic approach can be applied to provide the relationship:

$$\mathbf{p}\boldsymbol{K_a} + \mathbf{p}\boldsymbol{K_b} = \mathbf{p}\boldsymbol{K_w}$$

The pK_b may be found from the pK_a using the relationship:

$$pK_b = pK_w - pK_a$$

$$pK_b = 14 - 9 = 5$$

(Choice A) −5 results from a calculation error of either dividing K_a by K_w or subtracting pK_w from pK_a.

(Choice C) 9 is the pK_a value rather than the pK_b.

(Choice D) 14 is the pK_w value rather than the pK_b.

Educational objective:
The acid dissociation constant K_a and the base dissociation constant K_b of an acid and its conjugate base are related by the self-ionization constant of water K_w according to the relationship $K_aK_b = K_w$. This relationship can also be expressed using a logarithmic approach as pK_a + pK_b = pK_w.